Model-based head-pressure control and screw-fill estimation for a cold-feed rubber extruder

Systems engineering dossier. Pre-baseline draft, revision A, 10 Nov 2026.

This dossier follows the structure of C. S. Wasson, *System Engineering Analysis, Design, and Development* (2nd ed., Wiley 2016), cited as **[W]** with the printed page and then the PDF page.

The engineering work behind it is in this repository:

- the extruder model, identification, controllers and estimators in `src/xtrude/`;
- the PLC code in `plc/`;
- the study script, `scripts/run_study.py`, which writes `results/study.md`;
- the trade study, `scripts/aoa.py`, which writes `results/aoa.md`.

Every number in this dossier is reproduced by those scripts. Section 8.2 lists which numbers are **sourced** and which are **assumed**. Reading notes on the Wasson chapters used are in `docs/sources/`.

> **Status.** Everything is simulation plus a TIA Portal V18 compile. No data from a real line has been used. The dossier is a draft for a requirements review. The open items that need the plant are collected in section 12 (the TBD register).

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0. Summary of findings

All numbers in this summary are from simulation (§8). The closed-loop figures come from 24 random production scenarios of 1200 s each, using the same scenarios for every alternative and paired comparisons. Uncertainties are ±2 standard deviations across runs.

1. **What the pressure sits in front of decides what "pressure control" can do (§8.3).**
    - **Open power-law die:** the steady-state gain from speed to pressure is close to zero and its sign is uncertain. Viscous heating cancels the flow increase.
    - **Gear pump (the primary case):** the plant is lag-dominated with a slow pole at about 11 s. Its gain spans **6.7–110 bar/rpm** across operating pressures of 40–80 bar. Compound viscosity hardly changes the gain, because a stiffer batch heats itself and thins.
2. **A model-based RST controller does not regulate better than a robustly tuned PID.** Both were tuned on the same 10 plant variants under the same constraints (§10.1).
    - **Disturbance rejection:** worst-case IAE after a feed-loss step is **8.25 (RST) vs 8.83 (PID)**.
    - **Nonlinear Monte Carlo:** the 99th-percentile pressure deviation is **1.09 vs 1.05 bar**. RST is better in only 7 of 24 runs.
    - **What the RST does give:**
        - more robustness margin: worst Ms **1.46 vs 1.60**;
        - **12.7 % less speed travel**, better in all 24 runs;
        - **zero overshoot** on a 4-bar setpoint step, against **1.66 bar** for the PID.

      A PID with a setpoint prefilter also reaches zero overshoot.
    - **What limits both:** the 0.5 Hz flight-passing pressure ripple. Allowing 0.2 rpm of speed ripple instead of 0.1 rpm halves the IAE of both controllers. A speed-tracking notch makes both worse, because the ripple sits only 3–5 times above crossover.

- **MPC (DMC with an output-disturbance model): 43 % worse** at the 99th percentile. This is the known DMC limit for input disturbances on lag-dominated plants [C6].

3. **The measurable gain comes from topic 3.** Feed changes reach the pressure about 35 s after they reach the feed, and the drive torque sees them first. The fill estimator turns that into a feedforward speed. Results, paired against the same scenarios:

Measure	PID+FF vs PID	RST+FF vs RST
99th-percentile deviation	**−26 %** median (1.05 → 0.77 bar), better in 23/24 runs	**−27 %**, better in 21/24 runs
Time outside ±3 bar	halved	—
Actuator travel	+15–18 %	+15–18 %

4. **Compound left in the screw (topic 3), feasibility.** RMSE is over a content of about 8 kg.

Estimator	In range	Viscosity outside training range
Physics EKF (torque + head pressure + batch-change events)	**0.11 ± 0.09 kg** (1.4 %)	**0.16 kg**
Physics EKF, no batch events	0.29 kg	—
Learned model on torque, power and rpm only (the literal topic)	**0.29 ± 0.26 kg** (MLP)	0.38 kg
Learned model, all signals	0.09 kg	—
Static speed/gear-pump mass balance	0.55 kg	—

   - **Batch events:** without them the EKF error rises to 0.29 kg, because a viscosity change is then misread as a feed change.
   - **Learned models need labels.** Their training targets were the simulator's true mass. On a real line every label is a weighed purge, while the EKF needs none.
   - **Static mass balance:** it fails in run-out, with an error of 3.6 kg at the starvation point.

   Verdict. Estimating the screw content is **feasible**. From torque, power and rpm alone, a learned model reaches about ±0.5 kg (2σ), only within the trained compound range, and only once labelled purges exist. The recommended route is the physics EKF using the head pressure the loop already measures. It is the most accurate and robust option that needs no labels.

   Weak point. At the end of a run-out the EKF overestimates the last 0.2–0.35 kg by **0.43 kg on average** (max 0.62). This misses SPS-312 in 3 of 12 run-outs; the learned models are better there.

5. **Fits the PLC.** The worst-case added execution is **1.65 ms per 100 ms cycle** at datasheet instruction times. The EKF filter call is 1.60 ms. With a 3× margin this is 4.95 ms, within the 5 ms allocation.
   - All four SCL blocks compile in TIA Portal V18 with **0 errors, 0 warnings**.
   - They match the Python references in float32.
   - Float32 broke the RST's static-gain identity and caused an offset of several bar. The SCL now enforces T(1) = R(1) and S(1) = 0 in its code structure (§9, F10).

6. **Recommendation (§10.3).**
   - Build the fill estimator and its early feed-loss warning.
   - Re-identify and robustly retune the existing PID, adding a setpoint filter.
   - Offer the estimator feedforward and the RST as line-trial candidates (VT-04, VT-05).
   - The line trial, not this model, decides whether the RST's margin and wear advantages are worth its maintenance cost.

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1. Problem and stakeholders ([W] ch. 4–5)

1.1 Problem statement

The problem statement follows the guidelines in [W] p.92/122: one sentence, no cause, no solution, no blame.

> During start-up, compound changes and steady production on the cold-feed extruder line, the pressure in front of the outlet pump drifts outside its working window. This produces extrudate that is off gauge or has to be purged as scrap.

Discussion. This is kept separate from the statement itself, as the guidelines require.

The fair topic is phrased as a solution: "replace the current PID … by polynomial or matrix control" (topic 2), plus "predict the material left in the screws with NN/AI" (topic 3). [W] warns that stakeholders often state a need as a solution (p.103/133), and recommends the Five Whys (p.104–105).

Applied to topic 2, the Five Whys run as follows:

1. **Why replace the PID?** Because the pressure is not held well enough.
2. **Why does that matter?** Pressure in front of a gear pump decides how well the pump fills. That fill sets the output per revolution, and the output sets the gauge.
3. **Why is it not held?** Mainly because of disturbances in the feed strip, and a feedback controller sees them only after they have travelled down the screw.
4. **Why does that matter for the choice of controller?** Section 10 shows that a better *feedback* law buys little. The lever is *information about the feed earlier*.
5. **Where does that information come from?** From the screw-fill estimator, which is topic 3.

This is why the two topics are one project. The quantity topic 3 wants, the compound in the screw, is also the state that topic 2 needs as a measured disturbance.

1.2 Stakeholders ([W] Table 4.2, p.84–85/114–115)

Role ([W] Table 4.2)	Who on the line	What they need from this system
User / End user	extruder operator	Stable pressure without retuning. Fewer alarms. A clear mode display. The screw content shown during run-out.
System owner	production / process engineering	Less scrap per shift. Less start-up waste. Smaller gauge spread.
Maintainer	automation / maintenance engineers	Code they can read in TIA Portal. Re-tuning from a short test. A clear response to sensor faults.
Customer's customer ([W] P4.3, p.78)	tyre building / quality	Constant extrudate weight per metre.
Developer	thesis student + supervisor	A design that can be verified and validated within the thesis.
Independent test agent ([W] P13.15, p.289/319)	the plant's process engineer on the line trial	Acceptance against the measures of effectiveness (MOEs), independent of the developer.

1.3 Needs, measures of effectiveness and measures of performance ([W] p.100–101/130–131, Fig. 5.8 p.116/146)

ID	Need (stakeholder language)	MOE / MOS	Contributing MOPs (section 5)
N1	Less off-gauge scrap in steady production	MOE-1: off-gauge scrap kg per shift	SPS-301, SPS-302, SPS-305

ID	Need (stakeholder language)	MOE / MOS	Contributing MOPs (section 5)
N2	Less waste at start-up and at compound changes	MOE-2: kg purged per start / per change	SPS-304, SPS-311, SPS-312
N3	Smaller gauge (weight-per-metre) spread	MOE-3: output standard deviation, %	SPS-301, SPS-303
N4	Know how much compound is left in the screw	MOE-4: purge mass predicted vs weighed, kg	SPS-311, SPS-312, SPS-313
N5	No new failure modes and nothing hard to maintain	MOS-1: hours of automatic mode lost to the new functions; MOS-2: re-tune time	SPS-321…323, SPS-364

[W] says each MOE needs two or more MOPs (p.438/468). The table meets that rule.

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2. Concept of operations ([W] ch. 6)

The structure follows Table 6.1, the ConOps outline in [W] p.140/170.

2.1 Mission and context

What the line does. A cold-feed rubber extruder takes in a strip of compound through a feed roll. Its screw conveys the compound and heats it by shear. A gear pump at the screw tip meters the compound into the die. The extrudate is taken off at line speed.

What this system adds. It sits in the line PLC (S7-1500, TIA Portal) and has three jobs:

1. Hold the pressure at the gear-pump inlet by setting the screw speed.
2. Estimate the compound in the screw and show it.
3. Fall back safely when a sensor fails.

What it does not do. It does not replace the line's safety functions: the burst disc, the hard-wired overpressure switch and the emergency stop.

Context. The interfaces are listed in section 6.

Signal	Direction	Interface
Head-pressure transducer	In	IF-01
Drive speed and torque	In	IF-02 (PROFINET telegram)
Speed setpoint to the drive	Out	IF-02 (PROFINET telegram)
Melt thermocouple	In	IF-03
Operator commands	In	IF-04 (HMI)
Operator display	Out	IF-04 (HMI)
Gear-pump speed and feed status	In	IF-05 (line)
Batch-change event	In	IF-06 (line production records)

2.2 Phases and use cases ([W] P5.7 p.110/140; UC principles p.120–122/150–152; Table 6.4 p.143/173)

Phase	Use case (verb + outcome)	Trigger	Main success scenario	Alternate / failure scenarios
Pre-mission	**UC1** Identify the pressure dynamics	New compound family, new die, or maintenance	Technician runs the 2-minute ±0.3 rpm test. The tool fits a model and proposes RST/PID parameters.	Fit below 50 % → repeat the test. Model outside the robustness set → keep the old parameters.
Pre-mission	**UC2** Start up the line	Operator selects Auto after feed-on	The pressure reaches the window. The estimator converges (σ of mass < 0.3 kg). Mode RST+FF is enabled.	Estimator not converged → RST without FF.

Phase	Use case (verb + outcome)	Trigger	Main success scenario	Alternate / failure scenarios
Mission	**UC3** Hold pressure in production	Continuous	Pressure is held within ±3 bar despite feed and temperature variation.	Feed strip breaks → early warning from torque, then low-pressure alarm.
Mission	**UC4** Change compound batch	Line production records report a new batch entering the feed	The estimator frees its viscosity state for one screw transit. FF stays on.	No batch event → the viscosity change is misread as a feed change (quantified in section 9).
Post-mission	**UC5** Run out at end of campaign	Feed stopped, screw kept turning	The HMI shows the compound left in the screw. The operator stops when the screw is empty enough.	Pressure falls below the gear-pump fill limit → the gear pump starves. That is the point the estimate must be good at.
Any	**UC6** Recover from a sensor fault	Supervisor detects a fault	Speed is held or FF is disabled. An alarm is raised and acknowledged.	Repeated faults → maintenance call (section 9).

Coverage check. [W] Fig. 6.3 (p.137–138) is a matrix of operations against phases. Here it reduces to this: UC1 is maintenance-only, UC2–UC4 are production, and UC5 is post-mission. The "Conduct Mission only" gap of Author's Note 6.1 (p.137/167) is closed by UC1, UC5 and UC6.

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3. Modes and states ([W] ch. 7)

Definitions are from [W] p.148–149/178–179. A **mode** is a user-selectable label that enables a set of capabilities. A **state** is an observable, measurable condition.

Table 3-1: modes. These are implemented in `FB_PressureSupervisor`.

#	Mode	Enabled capabilities	Selected by	Exit criteria ([W] Author's Note 7.4: define the current mode's exit criteria)
0	Manual	Operator speed. Estimator runs and displays.	Operator	Operator selects 1–3
1	PID	Baseline loop	Operator	Operator change, PFault → 4, OverPressure → 5
2	RST	Model-based loop	Operator, or automatic fallback from 3	Same as 1
3	RST+FF	Model-based loop + fill feedforward	Operator	TorqueFault or estimator invalid → 2; PFault → 4; OverPressure → 5
4	SpeedHold (degraded)	Last good speed held	Automatic	Fault cleared **and** acknowledged → requested mode
5	SafeStop (emergency)	Speed ramped to minimum	Automatic	Pressure below trip **and** acknowledged

State variables ([W] P7.2–7.3, p.154–156):

- **Operational state:** mode number, fault latches, estimator valid.
- **Dynamic state** ([W] P7.9, p.166/196): "batch in transit", the timer set by the batch event; and "run-out", with the feed off.

Allowable and prohibited actions ([W] P7.10 p.169/199, Fig. 7.14 p.171/201):

- Prohibited: entering 3 while the torque signal is invalid; entering 1–3 while PFault is latched.
- Not advisable: the identification test (UC1) during run-out.
- All transitions are bumpless. Controllers that are not selected track the applied speed. This is verified in `tests/test_scl.py` and requirement SPS-306.

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4. Requirements ([W] ch. 19–22)

The outline follows [W] Table 20.1 (p.417–418/447–448). Section 3.0 of the outline is the global requirement ([W] P22.2, p.448): the system shall meet these requirements before, during and after exposure to the operating conditions of §3.5.

Statements follow [W] P22.5 (p.449), one capability and one outcome each ([W] P22.9 p.451). They are labelled Threshold or Objective ([W] p.410/440). The verification methods are I, A, D and T ([W] p.270–272).

Provisional values. Some values are marked **(P)**. Each traces to an analysis in this repository ([W] P19.14, p.402/432) but has not been measured on the line. They are carried as TBR in the TBD register in section 12. [W] p.426/456 warns that a value backed by no analysis is not a requirement. Each provisional value names the analysis behind it.

4.1 Missions and operational performance (§3.1–3.2)

ID	Requirement	Threshold	Objective	Verif.	Trace
SPS-201	With the new control functions enabled, the line shall produce no more off-gauge scrap per shift than with the current controller.	≤ baseline	≤ 0.7 × baseline (TBD-1)	T (line trial, section 11)	MOE-1
SPS-202	Start-up waste per start shall not exceed the current value.	≤ baseline	≤ 0.8 × baseline (TBD-2)	T	MOE-2
SPS-203	The automatic modes (1–3) shall be available for at least the stated fraction of production time, excluding line stops.	≥ 99 %	≥ 99.8 %	A + T	MOS-1

4.2 Capabilities (§3.3)

ID	Requirement	Threshold	Objective	Verif.	Trace
SPS-301	In mode 3 during production with feed-efficiency changes up to ±8 %, the controller shall keep the head pressure within ±3 bar of setpoint.	≥ 99 % of time (P)	≥ 99.9 % (P)	A (MC, §8) + T	N1, N3
SPS-302	After a step feed loss equivalent to 1 rpm, the peak pressure deviation shall not exceed the stated value on every plant variant of §8.4.	3.5 bar (P)	2.5 bar (P)	A	N1
SPS-303	The loop shall keep a maximum sensitivity Ms ≤ 1.6 (modulus margin ≥ 0.625) on every plant variant of §8.4.	1.6	1.5	A	N3
SPS-304	On a 4 bar setpoint step, the pressure shall settle within ±0.5 bar in ≤ 20 s with an overshoot ≤ 1 bar.	20 s	12 s	A + T	N2
SPS-305	Speed ripple caused by the 1 % flight-passing pressure ripple shall not exceed 0.1 rpm amplitude. RMS speed noise from the pressure noise shall not exceed 0.1 rpm.	0.1 rpm	0.05 rpm	A + T	N1, N5
SPS-306	A mode change shall not step the speed setpoint by more than 0.2 rpm.	0.2 rpm	0.05 rpm	T (SCL test) + D	N5
SPS-311	In production, the estimate of the compound mass in the pumping section shall have an RMS error no larger than the stated value.	0.3 kg (≈ 4 %) (P)	0.15 kg (P)	A + T	N4
SPS-312	During run-out, at the moment the head pressure first falls below 25 bar, the estimate error shall not exceed the stated value.	0.5 kg (P)	0.25 kg (P)	A + T	N2, N4
SPS-313	The estimator shall report a 1-σ uncertainty with every estimate.	reported	—	I + A	N4
SPS-321	On a pressure-signal fault (channel status, out of range, or frozen for 20 s while the speed moves), the system shall switch to SpeedHold within one 100 ms cycle and raise a latched alarm.	100 ms	—	T	N5
SPS-322	On a torque-signal fault or an invalid estimator, the system shall disable feedforward within one cycle.	100 ms	—	T	N5
SPS-323	At head pressure ≥ the trip value, the system shall ramp the speed down at ≥ 5 rpm/s and latch an alarm. This is an operational function, not a safety function (§9).	5 rpm/s	—	T	N5

4.3 Interfaces (§3.4)

Only external interfaces are specified here ([W] p.420/450). The details are in the interface table in section 6.2 (IF-01 … IF-06).

4.4 Operating environment (§3.5)

The requirements apply across this range:

Parameter	Range	Notes
Screw speed	15–45 rpm	
Inlet pressure setpoint	40–80 bar	
Compound viscosity factor κ	0.8–1.25	Relative to the reference batch
Melt temperature	100–140 °C	
Pressure transducer noise	0.3 bar RMS	Specified in TBD-5
Flight ripple	1 %	Specified in TBD-5

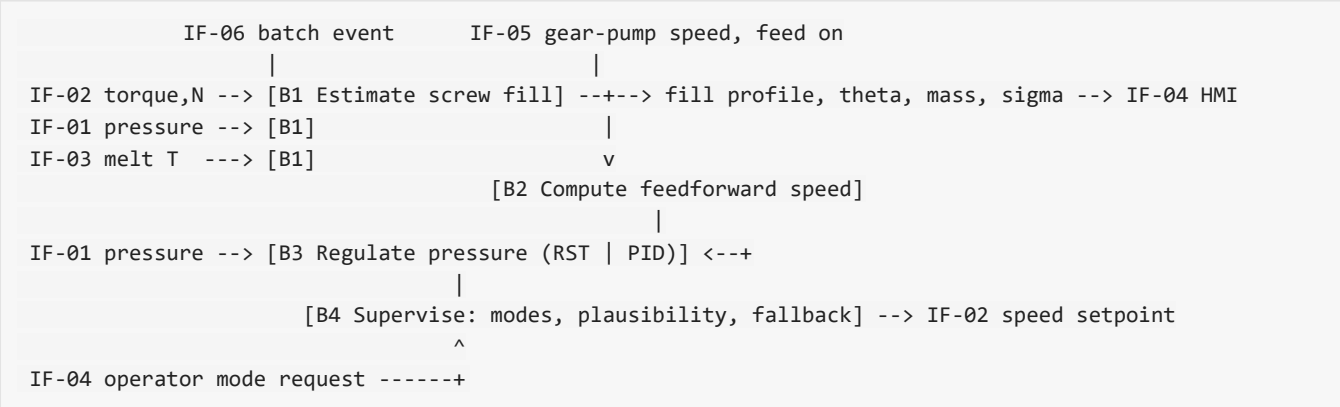
4.5 Design and construction constraints (§3.6)

ID	Constraint	Verif.
SPS-361	The added functions shall run in a cyclic-interrupt OB at 100 ms on an S7-1513-class CPU. Their worst-case execution shall be ≤ 5 ms, including the margin of §7.1.	A (§7.1) + T (on-target cycle statistics)
SPS-362	The code shall be SCL in TIA Portal V18 and compile with 0 errors and 0 warnings. No third-party libraries.	I + T (done, §11.2)
SPS-363	The functions shall use only signals already on the line: IF-01 … IF-06. No new sensors.	I
SPS-364	Re-identification shall need ≤ 2 min of line time at ≤ ±0.3 rpm speed excitation, during production.	D

5. Architecture ([W] ch. 26)

[W] p.551–552/581–582 uses four views: requirements, operations, behavioural and physical. "Logical architecture" is not Wasson's term; the behavioural view below plays that role.

Behavioural view. These are capabilities, not boxes ([W] P26.15, p.559/589).



Physical view. Each capability maps one-to-one to an SCL block. This keeps the traceability of [W] P21.2 (p.437/467): every leaf requirement is allocated to one entity.

Capability	SCL block	Requirements allocated
B1, B2	`FB_FillEKF` (estimator + `SpeedBalance` output)	SPS-311…313
B3	`FB_PressureRST`, `FB_PressurePID`	SPS-301…306
B4	`FB_PressureSupervisor`	SPS-306, SPS-321…323
Wiring	OB30 `Cyclic_Pressure` + `FC_LoadParams` + `LineIO` DB	SPS-361

Hardware. CPU 1513-1 PN (6ES7513-1AL02-0AB0, FW 2.9) as the reference target. The line PLC may be larger, which only adds margin.

Fault isolation and containment ([W] P26.16, p.559–560/589–590). A fault in B1 cannot reach the drive except through B2's feedforward, and B4 removes that path (SPS-322). A pressure fault removes B3 (SPS-321). There is deliberately no redundant transducer: [W] p.567 warns that redundancy through a shared connection is cancelled, and the plant already has a hard-wired overpressure protection layer.

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6. Interfaces ([W] ch. 27)

6.1 Ownership

The line PLC owns all interfaces ([W] P27.3, p.577/607). This project uses one system interface control document (ICD), which [W] p.582 advises for small projects. That ICD is the table below. Mnemonics follow [W] P27.6 (p.583).

6.2 Interface table

ID	Source → destination	Signal	Type / range	Rate	Integrity / fault handling
IF-01	Pressure transducer → AI card → PLC	Head pressure	4–20 mA, 0–500 bar (TBD-5)	Read every OB30 call	Channel status bit, range check, frozen-signal detection (SPS-321)
IF-02	Drive ↔ PLC (PROFINET telegram)	Speed setpoint out; actual speed and torque in	rpm; N·m	≤ 100 ms	Telegram valid bit; torque plausibility 0… 1.5 × rated
IF-03	Melt thermocouple → PLC	Melt temperature	°C	100 ms (sensor lag ~s)	Range check (TBD-6)
IF-04	HMI ↔ PLC	Mode request, acknowledge, setpoint; active mode, mass, σ, alarms	Int / Real / Bool	HMI cycle	—
IF-05	Line PLC → PLC	Gear-pump speed (relative), feed-roll running	Real, Bool	100 ms	Plausibility vs line speed
IF-06	Line production records → PLC	Batch-change event (new batch at the feed)	Bool, one cycle	Event	If missing: FF continues but section 9 quantifies the error. Treated as a dependency risk.

Timing (step 7 of [W] p.583/613). One OB30 call reads all inputs and writes the speed setpoint. The drive's speed loop is outside this system; its time constant enters the plant model as 0.3 s, with a variant at 0.6 s.

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7. Performance budgets and margins ([W] ch. 31)

7.1 PLC cycle-time budget ([W] P31.4 apportionment, p.668; §31.4 frame-based systems, p.679/709)

The operations come from running the SCL through the translator with operation counters (`src/xtrude/scl.py`). They are costed at the datasheet times of the CPU 1513-1 PN [P1]: 256 ns per Real operation, 64 ns per integer operation, 48 ns per move, and 20 × 256 ns per EXP or LN (assumed).

Allocation in the 100 ms OB30 frame	Count-based estimate	Allocated (estimate × 3 margin, [W] p.669)
FB_FillEKF (filter call, every 5th cycle: 3781 Real ops, 7251 integer/loop, 3009 moves, 5 EXP/LN)	1603 µs	4.81 ms
FB_PressureRST (103 Real ops)	35 µs	0.11 ms
FB_PressurePID (runs instead of RST)	9 µs	—
FB_PressureSupervisor	3 µs	0.01 ms

Allocation in the 100 ms OB30 frame	Count-based estimate	Allocated (estimate × 3 margin, [W] p.669)
Total of the new functions	1650 µs (1.6 % of the cycle)	**4.95 ms ≤ 5 ms (SPS-361)**

The allocation is tight because of the EKF. If the on-target measurement (VT-07) exceeds it, the covariance update can be split over the 5 calls between filter steps, which divides the peak by about 5. Nothing else needs to change. | Reserve kept at system level ([W] Fig. 31.1, p.670/700) | | ≥ 95 ms for the rest of the line program |

Why a 3× margin. The datasheet figures are typical times for simple instructions. Indexed access into optimized DBs and FB-call overhead are not in them. [W] p.669 gives 2–3× as the usual engineering margin. **Verification:** read the OB30 execution time from the CPU's cycle statistics during the line trial (§11, VT-07). This budget is a TPM ([W] P21.4, p.442/472): the prediction now becomes the measured value on the target.

7.2 Pressure → gauge error budget ([W] P31.5 synthesis by model, p.669/699)

The gear pump's fill curve couples the inlet pressure to the output. Under the assumed fill curve, Q ∝ 1 − e^(−P/15 bar), the sensitivity at 60 bar is about 0.12 % of output per bar (TBD-4).

Contributor	1σ at the pressure	→ output (gauge)
Closed-loop pressure deviation (PID 0.45, PID+FF 0.38, RST 0.46, RST+FF 0.49 bar std; Monte Carlo)	0.38–0.49 bar	0.045–0.058 % (simulated output std)
Transducer accuracy (TBD-5, e.g. ±0.5 % FS of 500 bar = 2.5 bar, *static* offset)	shifts the operating point, not the spread	≈ 0.3 % static shift, removed by the weight-per-metre trim
Speed ripple from flight ripple (SPS-305)	≤ 0.1 rpm → < 0.2 bar	< 0.03 %

Conclusion. In this configuration the pressure loop contributes at most about 0.1 % to gauge spread. Most of its value is in keeping the gear pump out of starvation, where efficiency collapses below about 25 bar, and away from overpressure. This is why SPS-301 is written as a *window* requirement and not as a variance requirement.

7.3 Estimator error budget

Contributor	Effect on mass estimate
Unknown viscosity of a new batch (no batch event)	RMSE rises from 0.11 to 0.29 kg (§9, F7)
Rheology mismatch (estimator n, b vs truth: 0.25/0.010 vs 0.2226/0.0089)	absorbed by θ in steady state; transient error inside the SPS-311 figure
Gear-pump fill-curve error (+10 % P_fill)	biases the steady-state fill; inside the SPS-311 figure
Reduced resolution (12 vs 80 compartments)	smooths the run-out front; dominates the SPS-312 error (mean +0.43 kg at the starvation point)

8. Modelling and simulation ([W] ch. 33)

8.1 Intended use and fidelity ([W] P33.1, p.706/736; fidelity matrix, p.708/738)

Intended use. The model is used for three things:

1. to compare controller and estimator *alternatives* under the same disturbances;
2. to size the PLC budget;
3. to produce provisional MOP values.

It is **not** accredited to predict absolute pressures or masses on the real line. [W] p.704 defines accreditation as acceptance *for a specific purpose*; that purpose is the comparison only. Before any absolute number is trusted, the model must be validated with line data (§11, VT-01 and VT-02). [W] P33.4 (p.716/746): a model is only as valid as its validation.

Phenomenon	Fidelity here	Source / basis
Drag and pressure flow in the channel	Newtonian flat-plate with shape factors, power-law viscosity at the channel shear rate	[S1] ch. 7, [S2] §6.3
Partially filled (starved) screw, moving fill front	1-D advection of fill, fill-front mass balance	[S4], [S2] ch. 9
Rheology	Power law + exponential temperature shift	[S7] n, b, T0; K scaled (§8.2)
Viscous heating	Barrel-wall work, one lumped temperature node	[S2] §6.3 power; lumped node assumed
Head / outlet	Compressible head volume; gear pump with fill curve, or power-law die	[S8] eq. 1 form for the die; gear-pump curve assumed
Drive	First order 0.3 s + 5 rpm/s ramp	assumed
Batch viscosity	Carried with the compound	assumed (physical)
Sensors	0.3 bar noise, 0.1 bar resolution, 1 % flight ripple; torque noise 0.5 % rated	assumed; ripple at screw frequency per [S8] (pressure oscillation observed)
Wall slip, yield stress, die swell, cure	**not modelled**	[S3], [S4] report slip / yield for rubber: a known gap

8.2 Sourced vs assumed parameters

Parameter	Value	Status
Power-law index n	0.2226	**sourced** [S7]
Temperature coefficient b	0.0089 /K, T0 = 90 °C	**sourced** [S7]
Consistency K	0.988e6 × 0.4 Pa·s^n	**sourced × assumed scale**. The scale of 0.4 makes nominal torque 65 % of the rating of the vendor drive class [S5]. Units are assumed SI.
Screw D, speed range, drive power	150 mm, ≤ 45 rpm, 185 kW	**sourced** (vendor class) [S5]
Lead, depth, flight width, clearance, pumping length	150, 18, 15, 0.5 mm; 1.2 m	**assumed**
Heat transfer UA	1100 W/K	**assumed**, calibrated to about 120 °C at the design point
Head volume, effective bulk modulus	4 L, 0.5 GPa	**assumed**
Gear-pump fill pressure	15 bar	**assumed (TBD-4)**
Fill at the feed opening	0.75 of drag capacity	**assumed**
Disturbance magnitudes	feed ±8 %, κ 0.8–1.25, line speed ±10 %, barrel ±5 K	**assumed (TBD-3)**

Plausibility checks (results/study.md §1):

- Design point: 741 kg/h at 30 rpm. At 45 rpm the model gives 1111 kg/h, against the vendor class's 1100 kg/h at maximum speed [S5].
- Specific energy 0.119 kWh/kg. The published range for a 40 mm rubber extruder is 0.3–1.7 kWh/kg [S6], and large machines sit lower.
- Torque 66 % of rated.
- Melt temperature 120 °C (103–132 °C over 15–45 rpm).

8.3 Why the outlet type matters (finding)

The fair topic does not say what the pressure sits in front of. `docs/img/outlet_step_response.png` compares the two outlet types:

- **Power-law die.** A 10 % speed step raises pressure by only about 2.5 bar. The extra viscous heat then lowers the die viscosity, and the pressure ends *below* its start. The steady-state gain is close to zero and its sign is uncertain. A pressure loop on an open die regulates the wrong variable for gauge.

- **Gear pump (the primary case here).** The same step raises the pressure steadily. The plant is lag-dominated with a slow pole near 10 s, and its gain varies about 16× with the operating pressure (§8.4).

Open question. Which configuration the plant has is open question Q1 (§12).

8.4 Plant variants (uncertainty set for design)

The variants are noise-free small-signal models identified at:

- nominal;
- κ 0.7 and 1.3;
- 40 and 80 bar;
- 20 and 40 rpm;
- bulk modulus 0.25 and 1.0 GPa;
- drive time constant 0.6 s.

DC gains range from 6.7 (at 40 bar) to 110 (at 80 bar) bar/rpm. The operating pressure dominates, through the gear-pump fill curve. Viscosity hardly matters (27.5 bar/rpm at κ = 0.7, 1.0 and 1.3), because a stiffer batch heats more and thins itself.

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9. Reliability, failure modes and the FMEA ([W] ch. 34)

9.1 What counts as a failure

[W] P34.3 (p.729/759) asks that this be defined and agreed with the stakeholders. **A failure of the new functions** is any of these:

- the head pressure leaves ±5 bar of setpoint for more than 10 s while in an automatic mode, with no line disturbance outside §4.4;
- a mass estimate is outside its own ±3σ for more than 60 s;
- an automatic mode is lost for a reason other than a detected sensor fault.

Drift counts as a failure ([W] p.728).

9.2 FMEA

The FMEA takes its columns from [W] Fig. 34.20 (p.766/796). Each row is scored for severity S, occurrence O and detection D, each from 1 to 10. The risk priority number is RPN = S × O × D, after Dhillon ([W] Eq. 34.41, p.765/795). Occurrence values are engineering estimates (TBD-7). [W] P34.12 (p.740/770): RMA figures are estimates.

ID	Item / function	Failure mode	Local effect	End effect	Detection	Compensating provision	S	O	D	RPN
F1	Pressure transducer (IF-01)	Wire break / channel fault	Pv invalid	Wrong speed → starve or overpressure	AI channel status	SpeedHold + alarm (SPS-321)	8	3	1	24
F2	Pressure transducer	Frozen signal	Loop open	Pressure drifts unseen	No change for 20 s while speed moves	SpeedHold (SPS-321), `test_frozen_pressure_detected`	8	2	3	48

ID	Item / function	Failure mode	Local effect	End effect	Detection	Compensating provision	S	O	D	RPN
F3	Pressure transducer	Slow zero drift (temperature, diaphragm)	Controlled to a wrong pressure	Gear pump fill shifts; gauge shift ≈ 0.12 %/bar	**Not** by plausibility. Zero check at the start of each campaign (screw empty, at temperature). The EKF pressure innovation trend is the condition-monitoring flag ([W] P-F curve, Fig. 34.25 p.787/817).	Recalibration procedure	5	4	6	**120**
F4	Pressure transducer	Noisy / damaged diaphragm	Speed noise	Gauge noise, drive wear	RMS of innovation > limit	Alarm; PID fallback with lower gain	4	2	4	32
F5	Drive torque (IF-02)	Telegram loss / implausible	Estimator blind	FF wrong	Telegram valid bit, range	FF off → mode 2 (SPS-322)	4	3	1	12
F6	Drive torque	Scaling changed (drive parameter edit)	θ estimate jumps	Mass estimate biased	θ outside the plausible band after transit	Alarm "torque scaling", FF off	4	2	5	40
F7	Line records (IF-06)	Batch event missing	Viscosity read as feed change	FF pushes the wrong way during transit	Only afterwards (pressure innovation)	Estimator error quantified (0.29 kg without events vs 0.11 kg with them); request the event as a dependency (Q4)	5	5	7	**175**
F8	Gear-pump speed (IF-05)	Wrong scaling	Wrong balance speed	FF bias	Feedback integrator absorbs it; FF deviation > 3 rpm alarm	FF uses *changes* only (OB30)	3	2	4	24
F9	Feed strip	Break / end of strip	Fill falls at the feed	Pressure collapses 30–40 s later	**Early** via torque (EKF), then low-pressure alarm	Operator warning before the pressure drops	6	4	2	48
F10	RST parameters	Wrong or corrupt download	Offset or instability	Pressure off target	Startup check that T(1) = R(1) and S(1) = 0 (see note)	Fall back to PID	7	2	3	42
F11	OB30	Cycle overrun	Time error OB80	Late setpoints	CPU diagnostics	Budget §7.1 (≤ 5 ms of 100 ms)	6	1	2	12
F12	Supervisor	Overpressure not caught by software	—	Mechanical damage	Hard-wired switch / burst disc (outside scope)	Independent safety layer	9	1	2	18

Note on F10. In float32 the rounded coefficients break both T(1) = R(1) and S(1) = 0, which causes a static offset of several bar. FB_PressureRST therefore enforces both identities in its code structure. This is verified by `test_rst_scl_matches_reference_in_closed_loop`.

Highest RPNs.

- **F7 (missing batch event).** Measured mitigation: the EKF error without batch events is 0.29 kg, against 0.11 kg with them.
- **F3 (transducer drift).** Mitigation: a calibration procedure, and the EKF innovation trend used for condition-based maintenance.

9.3 Availability

SPS-203 asks for the automatic modes to be available at least 99 % of production time. The inherent availability is Ai = MTBF / (MTBF + MCMT) ([W] Eq. 34.53, p.781/811). It is dominated by the transducer: F1–F4 are the faults that force SpeedHold. All values below are TBD-7.

Quantity	Assumed value	Basis
Transducer faults forcing SpeedHold (MTBF)	1 per 2 years ≈ 8000 h	Order of magnitude for melt-pressure transducers in rubber
Corrective time (MCMT)	1 h	Replace and zero the transducer
Resulting Ai	≈ 0.99988	From the two values above

The estimate must name its five criteria ([W] p.741/771): operating environment, mission duration, operating condition, elapsed time and probability of success. This is part of TBD-7.

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10. Analysis of alternatives ([W] ch. 32)

The method is the 12 steps of [W] p.689–690/719–720:

- normalised 100-point weights ([W] Fig. 32.8);
- linear utility functions ([W] Fig. 32.9);
- a ±10 % weight sensitivity check ([W] §32.7, p.696/726).

The full arithmetic is in `scripts/aoa.py`, and its output is `results/aoa.md`.

10.1 Pressure controller

All candidates were tuned on the same 10-variant set under the same constraints (§4.2: SPS-303 and SPS-305). The baseline is therefore a *robustly tuned* PID, not a strawman.

The factors and weights come from the stakeholder needs in §1:

Factor	Weight	Basis
Regulation	30	N1, N3
Robustness	15	SPS-303
Degradation behaviour	10	N5
PLC load	10	SPS-361
Maintainability	20	Maintainer, [W] Table 4.2
Implementation risk	15	—

The utilities are linear; the ranges and the reason for each judgement score are written in `results/aoa.md`.

Alternative	Regulation (p99)	Robustness (worst Ms)	Degradation	PLC load	Maintainability	Impl. risk	FOM
PID (robust retune)	0.98 (1.05 bar)	0.57 (1.60)	0.60	1.00	1.00	1.00	**89.0**
PID + fill feedforward	1.00 (0.77 bar)	0.57 (1.60)	0.90	0.83	0.80	0.70	**82.4**
RST	0.97 (1.09 bar)	0.76 (1.46)	0.60	0.99	0.60	0.80	**80.5**
RST + fill feedforward	1.00 (0.84 bar)	0.76 (1.46)	0.90	0.83	0.50	0.60	**77.8**
MPC (DMC)	0.80 (1.60 bar)	0.30 (n/a)	0.60	0.75	0.30	0.40	**54.0**

Sensitivity ([W] §32.7). No single ±10 % weight change moves the winner.

Assumption tests ([W] P32.10). These do not re-score the study; they show what the ranking depends on.

- **Regulation utility.** The regulation range above, 4 bar → 0 and 1 bar → 1, flattens the regulation differences, because every alternative already meets the ±3 bar window. Stretch the utility to the observed range, 1.6 bar → 0 and 0.7 bar → 1, and **PID + fill feedforward** wins at every regulation weight from 30 to 60.
- **Actuator wear.** Adding a 10-point actuator-wear factor keeps PID first and moves RST to second.
- **What decides it.** The choice between PID and PID+FF depends on one stakeholder value: how much a ~26 % cut in pressure peaks is worth when gauge spread is already only ~0.05 %. That value has to come from the plant (TBD-1, TBD-3).
- **RST does not win under any test.** On this model its advantages are margin, wear and setpoint tracking, not regulation.

10.2 Fill estimator

The factors and weights:

Factor	Weight
Accuracy in range	35
Accuracy out of the trained viscosity range	25
Run-out accuracy	15
Sensors needed	10
Explainability to maintenance	15

Alternative	In range	Out of range	At run-out point	Sensors	Explainability	FOM
Physics EKF (torque + pressure + batch events)	0.112 kg	0.160 kg	0.43 kg	0.7	0.8	**81.5**
MLP, all signals	0.087 kg	0.148 kg	0.08 kg	0.7	0.2	**78.1**
Ridge, all signals	0.215 kg	0.295 kg	0.93 kg	0.7	0.5	59.0
MLP, torque/power/rpm (topic 3 literal)	0.293 kg	0.376 kg	0.12 kg	1.0	0.2	57.5
Ridge, torque/power/rpm	0.371 kg	0.492 kg	1.22 kg	1.0	0.5	40.0
Static mass balance	0.555 kg	0.533 kg	3.61 kg	1.0	1.0	31.0

Sensitivity. No single ±10 % weight change moves the winner.

What the table does not show: labels. The MLP almost ties with the EKF, but its accuracy was bought with training labels the simulator gave for free. On a line, labels cost one weighed purge each. For the same reason the learned models only beat the EKF at the run-out point because run-outs were in their training set. The plan therefore has three steps:

1. Deploy the EKF.
2. Log weighed purges (VT-02, VT-08).
3. Train a learned *residual* on top of the EKF once enough labels exist, combining the two instead of choosing one. This is listed as future work.

10.3 Recommendation

1. **Build the screw-fill EKF** (`FB_FillEKF`), together with the batch-change event from the line records (IF-06).
    - **First, in observation mode:** it shows the compound in the screw and warns of feed loss about 30 s before the pressure drops.
2. **Re-identify and robustly retune the existing pressure PID.** Use the 2-minute ±0.3 rpm test of UC1 and the 10-variant tuning of §8.4.
    - **Add a setpoint filter**, so start-up ramps do not overshoot (SPS-304).

- **Cost and benefit:** cheap, and it removes the dependence on manual tuning. 3. **Offer two candidates in the line trial**, each against the retuned PID (VT-04, VT-05):
- **PID or RST with fill feedforward:** expected −26 % at the 99th percentile of the pressure deviation, at the price of +15 % actuator travel.
- **RST alone:** expected equal regulation, Ms 1.46 instead of 1.60, −13 % actuator travel. 4. **Let the plant decide on topic 2 from its own data.** This study does not support the premise that polynomial control will regulate better than a well-tuned PID on this plant. It does show the polynomial design is **implementable, verified in float32 and compiled**, and where its real advantages are. Section 12 lists what would change this conclusion:
- an outlet with more dead time;
- stronger disturbances than assumed;
- a die instead of a gear pump.

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11. Verification and validation ([W] ch. 13, 28)

[W] P13.3 and P13.4 (p.275/305) separate two questions. **Verification** asks whether the system meets its requirements. **Validation** asks whether it is the right system for the need. Verification here uses the fewest and cheapest methods that are sufficient ([W] P13.11, p.283/313).

11.1 Requirements verification traceability matrix ([W] Table 22.3, p.458)

Req.	Method	Evidence so far	Status
SPS-301	A (MC) + T	results/study.md §4	A: threshold met by all five (worst 0.31 % of time outside ±3 bar); objective of 99.9 % not met (best PID+FF 99.83 %); T open
SPS-302	A	results/study.md §3	Threshold met (worst peak PID 2.95, RST 2.91 bar); objective of 2.5 bar not met
SPS-303	A	results/study.md §3	Met (PID 1.60, RST 1.46; RST meets the 1.5 objective)
SPS-304	A + T	results/study.md §3, 4-bar step columns	RST met (8.0 s, 0 bar); PID alone fails the overshoot (1.66 bar), PID with setpoint filter met (8.4 s, 0 bar); T open
SPS-305	A + T	results/study.md §3 (ripple, noise columns)	A met; T open
SPS-306	T + D	`tests/test_scl.py` (tracking on Enable = FALSE)	T met (SCL); D open (PLCSIM/HMI)
SPS-311	A + T	results/study.md §5	A: threshold met (EKF mean 0.112 kg, worst run 0.241 kg); objective of 0.15 kg met on the mean only; T open
SPS-312	A + T	results/study.md §5 run-out table	A: **not met**, error > 0.5 kg in 3 of 12 run-outs (max 0.62 kg); improvement listed in §13; T open
SPS-313	I + A	`MassSigma` output; ±2σ band in `scenario_timeline.png`	I met; A coverage statistic open
SPS-321	T	`test_supervisor_fallbacks`, `test_frozen_pressure_detected`	met (SCL in translator); PLCSIM open
SPS-322	T	`test_supervisor_fallbacks`	met (SCL); PLCSIM open
SPS-323	T	`test_supervisor_fallbacks`	met (SCL); PLCSIM open
SPS-361	A + T	§7.1 budget; on-target cycle statistics	A met; T open
SPS-362	I + T	results/tia_compile*.txt, docs/img/tia_FB_PressureRST.png	**Met** (9 Nov 2026): 0 errors, 0 warnings for all blocks, OB30 at 100 ms

Req.	Method	Evidence so far	Status
SPS-363	I	§6 interface table	met
SPS-364	D	`ident.step_test` (120 s, ±0.3 rpm)	A met; D open

11.2 What has been verified in this repository

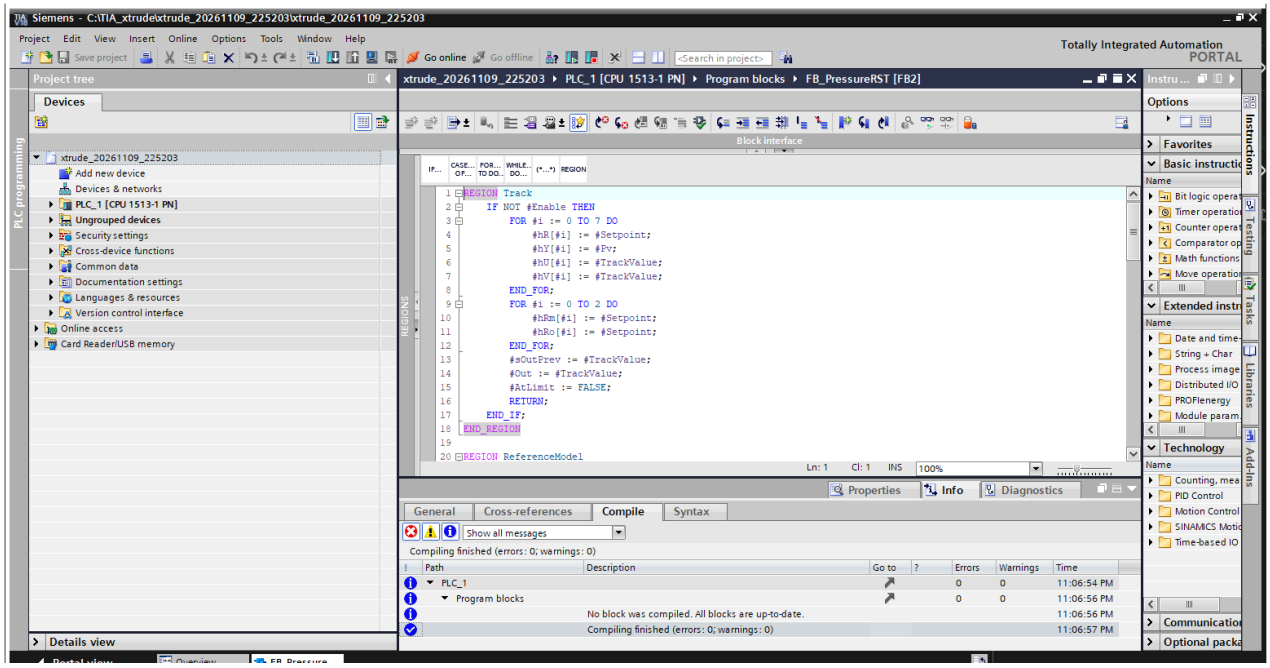
1. **SCL equivalence (V01).** The SCL that goes into TIA runs in CI through a translator that uses float32 Real:

   - PID: matches the reference to 1e-3 rpm;
   - RST: matches to 0.01 rpm in closed loop, with no static offset;
   - EKF: mass matches to 0.02 kg against the float64 reference.

   The translator is the author's own, and covers only the SCL subset used. TIA Portal is the authority.

2. **TIA Portal V18 compile (V02).** Done on 9 Nov 2026 through TIA Openness, using `scripts/tia_build.ps1` and `tia_ob30.ps1`.

   - **Target:** S7-1513-1 PN, FW 2.9.
   - **Compiled with 0 errors and 0 warnings:**
     - FB_PressurePID (FB1), FB_PressureRST (FB2), FB_FillEKF (FB3) and FB_PressureSupervisor (FB4);
     - FC_LoadParams (FC1);
     - the five DBs;
     - `Cyclic_Pressure` as a real OB30 cyclic interrupt at 100 ms.
   - **Evidence:** the logs are `results/tia_compile.txt` and `results/tia_compile_ob30.txt`. The screenshot is `docs/img/tia_FB_PressureRST.png`.



3. **Not yet: PLCSIM (V03), on-target timing, line trial.**

11.3 Line-trial plan (validation) ([W] P13.13, p.284/314; SITE, ch. 28)

The test environment builds up in stages ([W] Fig. 33.6, p.715): simulated, then PLCSIM with the plant model as an SCL FB, then the real line in observation mode, then the real line in closed loop. Integration is incremental, one capability at a time ([W] P28.1, p.605/635).

Step	Test case (phase → mode → UC, [W] Table 28.2 p.614/644)	Configuration	Acceptance criterion	Verifies
VT-01	Pre-mission → Manual → UC1	Real line, speed doublet ±0.3 rpm, PLC logs P, N, torque, T at 10 Hz	Fit ≥ 50 % on a second record; poles inside the variant set of §8.4	Model validity (§8.1)
VT-02	Mission → Manual (estimator only) → UC3	Observation mode: estimator runs, does not act; one purge weighed at the end	Weighed purge vs estimate within SPS-312	SPS-311/312 (first evidence)
VT-03	Mission → PID → UC3	Current PID retuned by the robust method of §8.4	Baseline numbers for SPS-201/202	Baseline
VT-04	Mission → RST → UC3	RST from VT-01's model	SPS-301, SPS-305 vs VT-03	SPS-301/305
VT-05	Mission → RST+FF → UC3, UC4	Batch event wired (IF-06)	SPS-301; no adverse effect at batch changes	SPS-301, F7
VT-06	Any → faults → UC6	Disconnect the transducer signal at the AI card in Manual with low speed; simulate a torque telegram loss	SPS-321/322 within one cycle	SPS-321…323
VT-07	Any	CPU cycle statistics over one shift	OB30 net time ≤ 5 ms	SPS-361
VT-08	Post-mission → RST+FF → UC5	Run-out, purge weighed	SPS-312	SPS-312
VAL-1	Validation over two weeks	A/B by shift (current vs new)	MOE-1…3 vs baseline	SPS-201/202, N1–N3

Rules for the trial.

- **Site order.** On site: read the state, take a backup, then make changes. Retests run in manual at slow speed first.
- **Independence.** The acceptance test is run by the plant's process engineer, not the developer ([W] P28.4, p.608/638; P13.15).
- **Configuration control.** Any change to the PLC code after verification invalidates that verification until it is re-run ([W] P13.16, p.289/319).
- **Regression.** Every parameter or code fix gets a regression run of `pytest` plus the affected VT steps ([W] p.609/639).

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12. TBD register and open questions

ID	Item	Needed for	How it gets resolved
Q1	Outlet type at the pressure sensor: gear pump, roller head, or open die?	Whole design (§8.3)	First meeting. It changes the plant class.
Q2	Current PID: structure, settings, cycle time, saturation and anti-windup method	Baseline (VT-03)	Read from the TIA project
Q3	Signals available: torque in the drive telegram? melt T? gear-pump speed?	SPS-363	I/O list
Q4	Is there a batch-change event (production records / barcode at the feed)?	F7, SPS-311	Production IT
TBD-1/2	Baseline scrap and start-up waste	SPS-201/202	Two weeks of production data
TBD-3	Real disturbance magnitudes (feed, κ spread from mixing-room Mooney data)	§4.4, MOPs marked (P)	Historian + lab data
TBD-4	Gear-pump volumetric-efficiency curve vs inlet pressure	§7.2	Pump datasheet
TBD-5	Transducer model, range, accuracy, noise	IF-01, §7.2	Data plate / datasheet
TBD-6	Melt temperature sensor position and lag	IF-03	Drawing

ID	Item	Needed for	How it gets resolved
TBD-7	Failure rates and repair times	§9	Maintenance records
TBD-8	Compound rheology (n, b, K) for the plant's compounds	§8.2	Capillary / RPA tests per ASTM D5099 [S9]

13. Limits of this work

- **Simulation only.** The model is built from textbook physics and one published compound. It is not validated on a line (§8.1).
- **Rubber-specific effects that are left out.** Wall slip and the yield stress of filled compounds [S3, S4] are not modelled. Both would change the filled length and the torque-to-fill relation.
- **The gear-pump curve is assumed.** The gauge budget (§7.2) depends on it directly.
- **The MPC tested is DMC with an output-disturbance model.** A state-space MPC with an input-disturbance model would do better ([C6]) but was not built. The AoA therefore rates MPC on what was tested.
- **The learned models were trained on simulated data.** Their out-of-range behaviour (§10.2) is the honest feasibility signal. Their in-range accuracy says little about real data.
- **The PLC timing is an estimate** from operation counts and datasheet figures. It is not a measurement.
- **The EKF misses SPS-312 at the end of a run-out.** It overestimates the last few hundred grams by 0.33–0.62 kg. The likely cause is the coarse 12-compartment grid smearing the emptying front. Candidate fixes, none of them tried:
    - a finer grid in the last third of the channel;
    - a dedicated run-out mode that tracks the empty front explicitly;
    - the learned residual of §10.2.
    - **Not every disturbance was varied.** The Monte Carlo covers feed, viscosity, line speed and barrel temperature. It does not cover transducer drift (F3) or a change of die or gear pump.

14. References

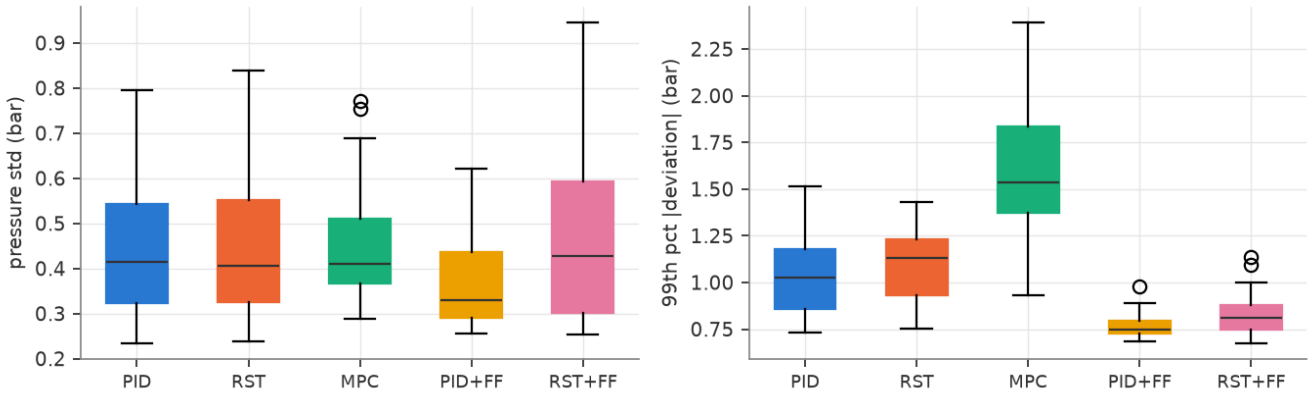
The full list, including the source status of each entry, is in `docs/sources/REFERENCES.md`. The main entries:

- **[W]** C. S. Wasson, *System Engineering Analysis, Design, and Development*, 2nd ed., Wiley, 2016.
- **[S1]** C. Rauwendaal, *Polymer Extrusion*, 5th ed., Hanser, 2014.
- **[S2]** Z. Tadmor, C. G. Gogos, *Principles of Polymer Processing*, 2nd ed., Wiley, 2006.
- **[S3]** J. L. White, *Rubber Processing*, Hanser, 1995.
- **[S4]** Brzoskowski et al., *Int. Polym. Process.* 2 (1987), doi:10.1515/ipp-1987-0004.
- **[S5]** Vendor data, 150 mm cold-feed extruder class.
- **[S6]** KGK, energy in rubber extrusion.
- **[S7]** Ha et al., KSME 2007 (compound w6043).
- **[S8]** Previdi, Savaresi, Panarotto, IFAC 2005; *Control Eng. Practice* 14 (2006).
- **[S9]** ASTM D5099.
- **[C1]** Åström & Wittenmark, *Computer-Controlled Systems*, 3rd ed., 1997.
- **[C2]** Landau & Zito, *Digital Control Systems*, 2006.
- **[C3]** Bobál et al., *Digital Self-tuning Controllers*, 2005.
- **[C4]** Åström & Hägglund, *Advanced PID Control*, 2006.
- **[C5]** Camacho & Bordons, *Model Predictive Control*, 2007.
- **[C6]** Lundström et al., *Comput. Chem. Eng.* 19 (1995).
- **[C7]** Ljung, *System Identification*, 1999.
- **[C8]** Simon, *Optimal State Estimation*, 2006.
- **[P1]** Siemens data sheet 6AG1513-1AL02-2AB0 (based on 6ES7513-1AL02-0AB0), 2023.

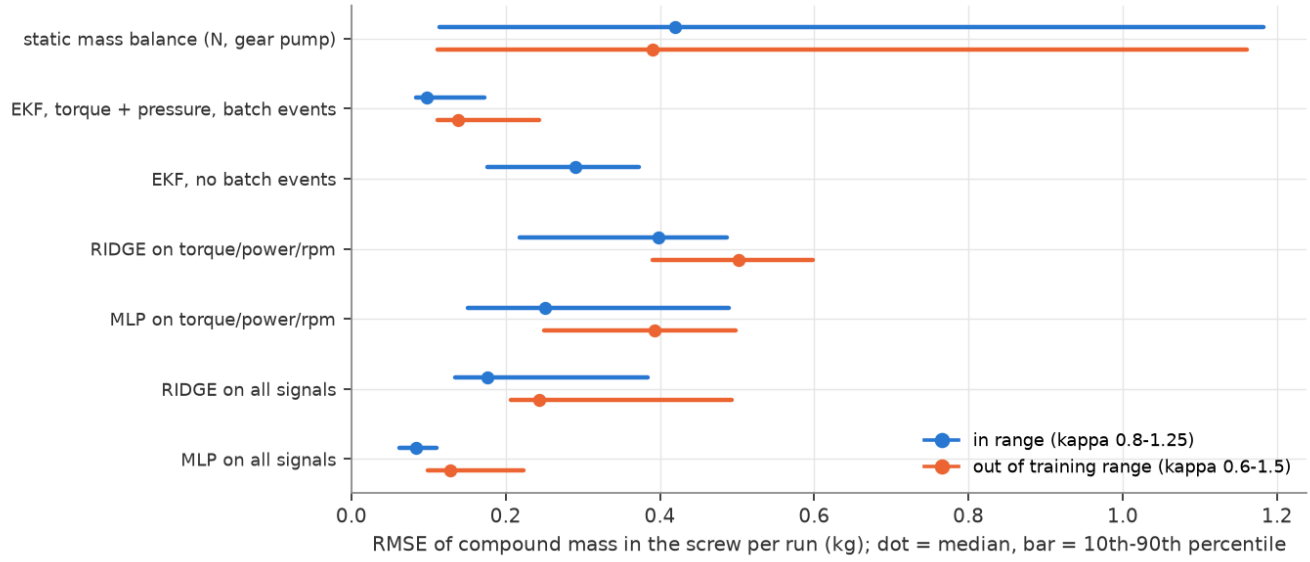
Appendix: figures

closed_loop_mc.png

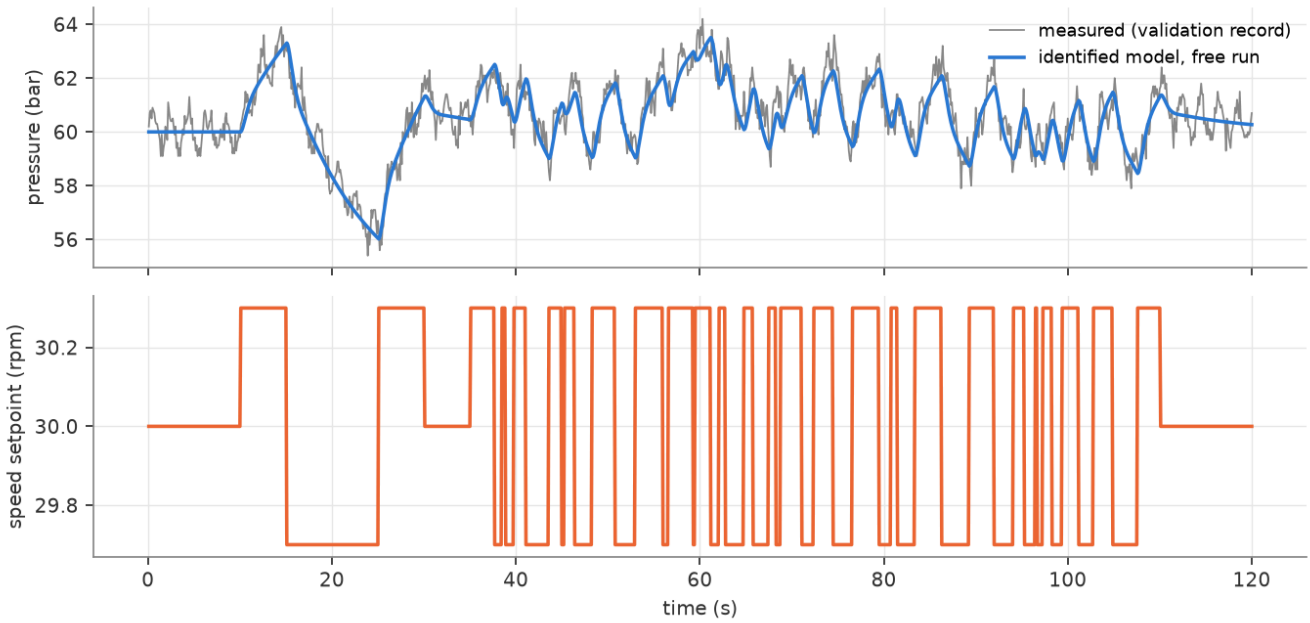
Closed loop, 24 random production scenarios x 1200 s



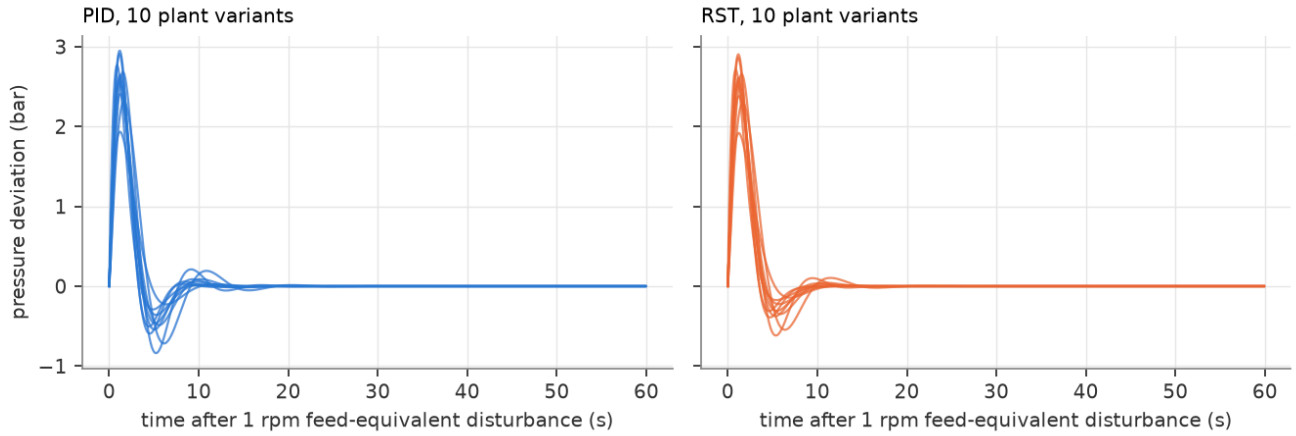
estimator_errors.png



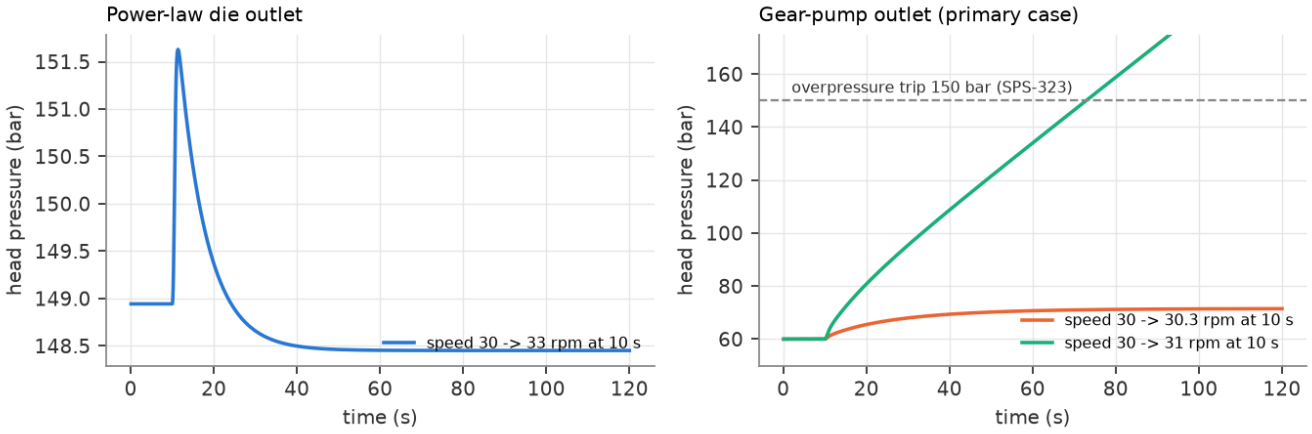
identification.png



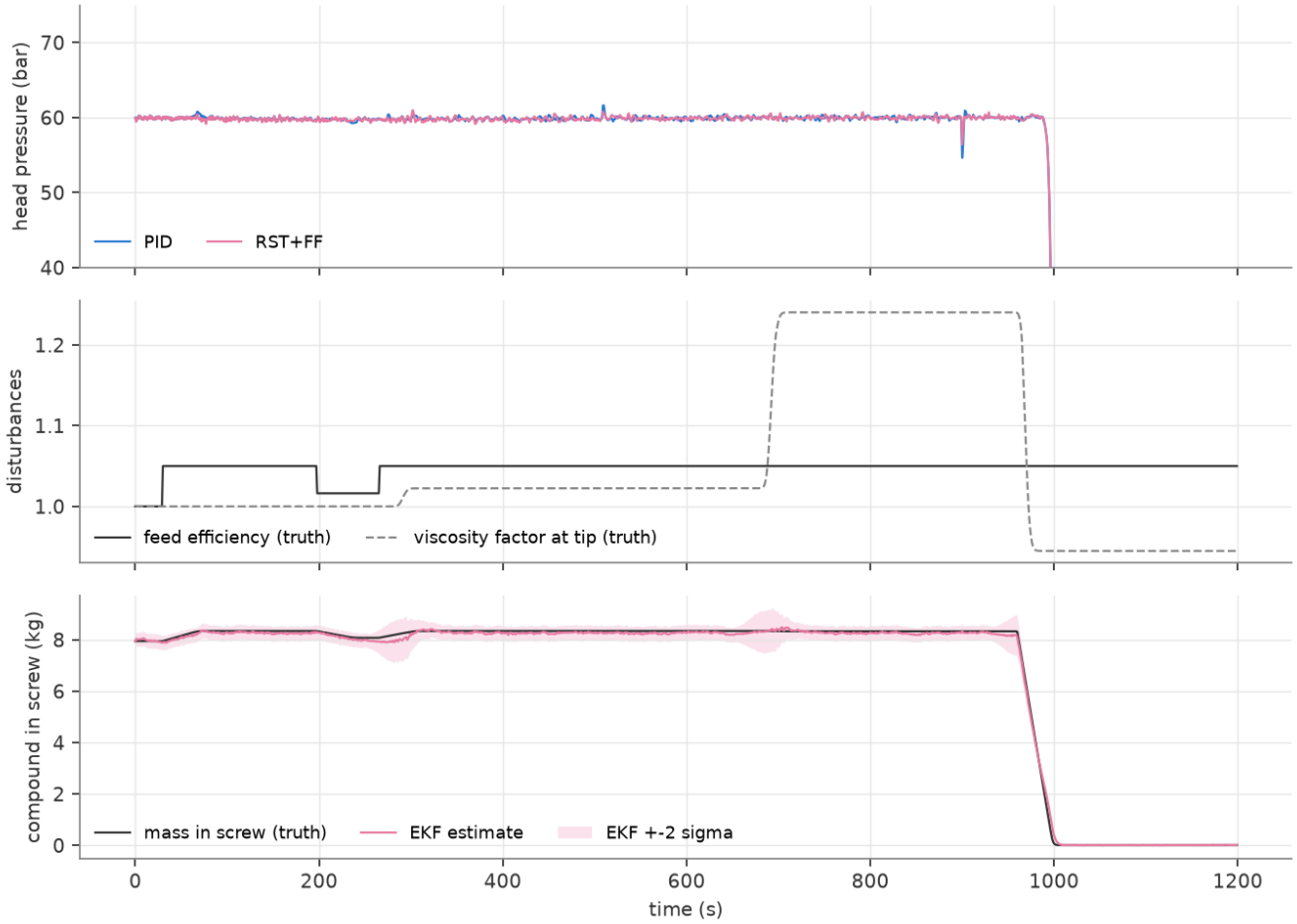
load_response_variants.png



outlet_step_response.png



scenario_timeline.png



tia_FB_PressureRST.png

